# JADE: Evaluating an Autonomous Trade Promotion Accountant

Planted-error evaluation, refusal on real data, and a walk-forward backtest on General Mills' 10-Ks

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October 2026

**Abstract**

JADE fills two finance seats at a consumer packaged goods manufacturer: the trade promotion accountant and the deduction analyst. It accrues promotional obligations on shipment as contra-revenue (ASC 606), matches retailer deductions to programs, clears what the agreement supports, disputes what it does not, restates closed months when terms change, and packages the close for a person to sign. Every number is computed by code into an insert-only, bitemporal ledger; a language model only drafts wording. We evaluate it three ways. *Planted errors:* 42 errors and 8 clean look-alikes hidden in 24 months of seeded books; JADE catches 42 of 42 with precision 1.000 at month 12, on the development seed and on two seeds it never saw. *Refusal on real data:* 4,996 real emails and 102 real commercial contracts; it acts on none. *Real books:* a walk-forward backtest of General Mills' accrued trade promotion liability, FY2021–FY2026, where the best estimator misses by 9.4% and the close's analytical review raises no false alarm on audited figures.

## I. Problem

Trade spend is the second-largest line on a consumer packaged goods income statement after cost of goods. General Mills sold $18.42B in fiscal 2026, 22% of it to Walmart, and carried $493M of accrued trade and coupon promotion liabilities at year end, which its 10-K names as a critical accounting estimate. The manufacturer books what it owes when the truck leaves and learns what it really owed months later, when the retailer deducts it from a payment, often at a different rate and with backup that does not tie. When the estimate trues up, the quarter moves: trade expense timing took 13 points off Q4 adjusted operating profit growth in FY2025 and added 7 points in FY2026.

Deduction-level data is never public, on either side of the trade. That shapes the evaluation: the transaction-level behaviour is tested on seeded books with an answer key (Sections V-A and V-B), refusal is tested on real public text (Section V-C), and the one estimate that is public, the year-end liability, is backtested on the real filings (Section V-D).

## II. System under test

Each step runs one month and handles each source row on the day it lands: read agreements into terms (each term cites its clause), accrue on shipment, match each deduction to a program, its shipments and its claim backup, run 12 controls mined from last year's decisions, then clear, dispute or ask. What the agreement cannot settle becomes a card for the person who owns the decision, with a recommendation; that person's answer becomes a precedent, reused for the same situation up to the amount approved. Five verifiability gates (journals balance, reversals mirror, no stale accruals, claims inside their windows, nothing cleared beyond what was accrued) block a period rather than warn. Segregation of duties is a capability check that throws: the person who negotiated a deal cannot approve its money.

## III. Evaluation method

### A. Seeded books and the answer key

A deterministic generator (seed 1337) produces 24 months of shipments, invoices, returns, deductions, claim backup, agreements and amendments as documents, and last year's human decisions, for General Mills' programs at Walmart, Costco and Kroger. Into these it plants 42 errors of known types (over-claims, duplicates, missing backup, stale accruals, an unentered amendment, self-approval attempts and others) and 8 clean look-alikes that must pass. The answer key (`data/answer_key.json`) is never visible to the system. Volumes are calibrated to the 10-K: every close checks the simulated liability against General Mills' reported ratio to net sales (2.16%–3.37% simulated, against 2.41% and 2.68% reported).

### B. Scoring

`tpa/evaluate.py` scores findings against the key by evidence: a finding counts against a planted row when its evidence names one of the row's specific ids (deduction, shipment, event, amendment or accrual key); program ids are too broad to count. A gate failure counts when the period's gate results carry a violation keyed to the row. We report, cumulatively by month:

- *precision*: findings that match a planted error, over all findings;
- *recall*: planted errors caught, over planted errors due by that month;
- *escalation rate*: findings in the month that went to a person;
- *autonomy*: customer-periods signed with no escalation raised in them.

The humans who answer cards in an unattended run are played: they take the agent's recommendation, or a model plays the role and must pick one of the card's options. Findings on rows injected live during a demo are scored apart, since the key never planted them.

### C. Generalisation and reliability

The same generator with seeds 42 and 2024, which were never developed against, places the same error types differently. This shows the system is not fitted to one answer key; it does not show it would catch an error type never planted. Reliability is measured as pass^$k$: the full year is run $k = 3$ times from the same seed and the ledger and decision hashes must be identical.

### D. Refusal on real data

An autonomous accountant's worst failure is acting on input it should not trust. We run real public text through the two readers that turn documents into source rows: 3,000 business emails (`corbt/enron-emails`) and 1,996 spam and ham emails (`SetFit/enron_spam`) through the mail reader, and 102 commercial contracts from CUAD (`chenghao/cuad_qa`), wrapped as amendments, through the agreement reader. The pass condition is zero actions and zero extracted terms. We also reworded 28 deduction notices by hand to test that the mail reader holds what it cannot parse instead of guessing.

## IV. Backtest method

The one trade promotion number that is public is the year-end liability. `tpa/backtest.py` walks forward one fiscal year-end close at a time over FY2021–FY2026, using figures quoted from General Mills' 10-K filings on SEC EDGAR (`tpa/filings.py`). At each close an estimator sees only the 10-Ks filed before that year's own 10-K, plus the year's net sales, which the company knows at its close; the estimate is then scored against the liability the 10-K reported. Four estimators:

- *carry forward*: last year's liability;
- *annual ratio*: last year's liability over net sales, times this year's net sales (the measure the close's analytical review uses);
- *Q4 ratio*: the same on fourth-quarter sales, since claims settle within months;
- *mean Q4 ratio*: the Q4 ratio averaged over every year public so far.

Two of the close's controls are replayed on the real books. Because those books were audited, any flag is a false alarm: the *analytical review* (the liability ratio against the prior 10-K's, at the policy tolerance of ±50%), and a *restatement check* (a year end stated differently a year later). This scores the top-down estimate and the controls at company level, never deduction matching.

## V. Results

### A. Planted errors

TABLE I
SELF-IMPROVING: PLANTED ERRORS, SCORED MONTHLY (SEED 1337)

| Month | Planted | Caught | Precision | Recall | Escalation | Autonomy |
|---|---|---|---|---|---|---|
| 1 | 4 | 4 | 1.000 | 1.000 | 0.400 | 0.667 |
| 3 | 10 | 10 | 1.000 | 1.000 | 0.250 | 0.667 |
| 5 | 21 | 21 | 1.000 | 1.000 | 0.333 | 0.600 |
| 6 | 27 | 27 | 1.000 | 1.000 | 0.143 | 0.611 |
| 7 | 30 | 30 | **0.857**† | 1.000 | 0.222 | 0.571 |
| 8 | 35 | 35 | 1.000 | 1.000 | 0.133 | 0.542 |
| 9 | 37 | 37 | 1.000 | 1.000 | 0.800 | 0.518 |
| 10 | 40 | 40 | 1.000 | 1.000 | 0.143 | 0.533 |
| 12 | 42 | 42 | 1.000 | 1.000 | 0.500 | 0.556 |

Cumulative. †Six July rate conflicts were raised before the planted amendment they trace to entered the scoring window. Autopilot without the judgment engine catches 40 of 42.

TABLE II
GENERALISATION AND RELIABILITY

| Run | Caught | Precision | Recall |
|---|---|---|---|
| Seed 1337 (development) | 42 / 42 | 1.000 | 1.000 |
| Seed 42 (unseen) | 42 / 42 | 1.000 | 1.000 |
| Seed 2024 (unseen) | 42 / 42 | 1.000 | 1.000 |

Reliability, pass^3 on seed 1337: 3 of 3 full runs identical (979 ledger lines, 75 findings, ledger hash `fa85730190eb4cc1`, decisions hash `9b39dd7e5a9f7ba3`).

The escalation rate does not fall steadily because new kinds of situations keep arriving through the year (a learned control in September, a new ambiguous clause); repeated situations do not escalate, as Table III shows.

### B. Self-learning

TABLE III
SELF-LEARNING: ANSWERS REUSED AS PRECEDENT

| Situation | Cards | Share |
|---|---|---|
| First-time question, asked a person | 17 | — |
| Repeat situation | 40 | 100.0% |
| settled silently by precedent | **37** | **92.5%** |
| asked again | 3 | 7.5% |
| *Why the three were asked again* | | |
| FN-0061 $42,413.95 vs. approved $41,319.56 | | over approval |
| FN-0072 $48,166.44 vs. approved $42,413.95 | | over approval |
| FN-0075 $58,315.83 vs. approved $48,166.44 | | over approval |

Seed 1337, twelve played months. A precedent covers the same situation up to the amount a person approved; a larger exposure goes back to a person.

### C. Self-escalating: refusal on real data

TABLE IV
SELF-ESCALATING: INPUTS IT MUST NOT ACT ON

| Dataset | Source | $n$ | Acted on | Held / refused |
|---|---|---|---|---|
| `corbt/enron-emails` | Hugging Face | 3,000 | **0** | 3,000 |
| `SetFit/enron_spam` | Hugging Face | 1,996 | **0** | 1,996 |
| CUAD contracts (`chenghao/cuad_qa`) | Hugging Face | 102 | 0 terms | 102 |
| Reworded deduction notices | ours | 28 | 25 parsed → 28 after 2 fixes | |

Rewording found two real parser gaps (bulleted labels, HTML-escaped senders), fixed in `tpa/mail.py`.

### D. Real books: the 10-K backtest

TABLE V
ESTIMATE AT EACH CLOSE AGAINST THE REPORTED LIABILITY ($M)

| FY | Reported | Carry fwd | Annual ratio | Q4 ratio | Mean Q4 | Review drift |
|---|---|---|---|---|---|---|
| 2021 | 508 | 471.0 | 484.4 | 424.2 | 424.2 | +4.9% |
| 2022 | 420 | 508.0 | 532.3 | 549.3 | 504.0 | −21.1% |
| 2023 | 394 | 420.0 | 444.4 | 431.9 | 489.5 | −11.3% |
| 2024 | 425 | 394.0 | 389.4 | 369.2 | 436.4 | +9.2% |
| 2025 | 470 | 425.0 | 417.1 | 410.8 | 419.6 | +12.7% |
| 2026 | 493 | 470.0 | 444.4 | 475.5 | 433.0 | +10.9% |

Each estimate made from filings public before that year's 10-K. Review drift is the year's liability-to-sales ratio against the prior year's.

TABLE VI
REAL BOOKS: GENERAL MILLS 10-K BACKTEST, SUMMARY

| Estimate at each close | MAE ($M) | Mean error | Bias | Worst year |
|---|---|---|---|---|
| Carry last year forward | **41.7** | **9.4%** | −0.2% | 21.0% |
| Annual ratio to net sales | 53.9 | 12.3% | +0.9% | 26.7% |
| Q4 ratio to net sales | 63.9 | 14.4% | −0.9% | 30.8% |
| Mean Q4 ratio | 64.2 | 14.4% | +1.2% | 24.2% |

Accrued trade and coupon promotions at each fiscal year end, FY2021–FY2026 (six closes, $394M–$508M). Analytical review at ±50%: 0 false alarms; largest drift −21.1% (FY2022). Restatements of a reported year end: none.

Two findings follow. First, the audited liability swings by up to 21.1% a year relative to sales, so a ±50% analytical review can never fail on these books; a tolerance near ±22% would be calibrated on the filings. Second, no top-down rule beats carrying last year forward, and scaling by sales does worse: the balance follows which programs run and how late retailers claim, which is why JADE accrues program by program rather than as a share of sales.

## VI. LIMITATIONS

Transaction-level results are on simulated books: the generator and the error types are ours, and unseen seeds vary placement, not kind. The played humans accept recommendations; real reviewers will disagree, and the precedent store has only been exercised on the situations it saw. The backtest scores six company-level year ends, not deduction matching, and FY2021's estimate rests on one prior filing. The tolerances used are company policy settings, not GAAP requirements.

## VII. Reproducing

`make db-up && make db-scratch && uv sync && make seed` 24 months, seed 1337
`make players` the played year and its scores (Tables I and III)
`uv run python scripts/evaluate.py --as-of month-1 --as-of month-9` scoring at chosen months
`make backtest` Tables V and VI, no database needed
`tectonic docs/research/jade-research.tex` this paper